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Optimizing process performance visibility through additional descriptive features in performance measurement

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***Abstract*—Measuring performance of business processes is necessary to access information on their efficacy. This enables successful optimization, reengineering and alignment on strategies. In general, measurement is done via numeric KPIs. Though, especially processes of a quantitative, non-deterministic or supportive nature are hard to describe via figures which leads to a reduced perception of performance problems. In this paper, the concept of visibility is presented. It fosters the development of a suitable Performance Assessment System (PAS) that broadens the range of indicators to e.g. goals, complexity, maturity, relations or dependencies in addition to KPIs to finally improve visibility of process performance.**

Keywords*— *Visibility, BPM, KPI, Non-numeric, Indirect, Descriptive, Generic, Indicator, Metrics, Goals, Semantics, Ontologies, Performance, Process Performance Management, Business Performance Measurement, Performance Measurement System, Performance Assessment System

I. INTRODUCTION

Business Process Management (BPM) is identified as a key factor to increase efficiency within operative business processes. In order to (re)-design, adapt, and execute business processes, their analysis is essential [52]. However, various empirical findings have identified a gap between analysis, implementation, and execution [37], [46] and a lack of knowledge concerning strategic management and top-down execution [29], [25]. One reason is the missing of methodological guidance of Performance Measurement Systems (PMS) to assess and manage business processes.

PMS have become important and critical part to measure the success of business processes and to give management information. They rely mostly on numeric oriented key performance indicators (KPI) [50], [46] which are strongly related to highly specified, formalized, and automated business processes [9]. However, KPI supply the management with simplified and comprised measures. The reduced meaning of KPIs does not allow process assessment of non-measureable success factors which means that relevant information is left out. Especially processes with non-numeric outcome, i.e. indirect processes like Human Resources or Public Relations that are related to qualitative outcomes and hard to measure [1], [31], [32] are lost for review and control through KPIs. So, neither success nor improvements of qualitative business processes are assessable.

Though some PMS consider qualitative key figures to measure processes, there is a lack of comprehensive qualitative performance [56], [50]. Lynch and Cross [34] e.g. provide different perspectives within a Performance Pyramid by combining non-monetary and customer viewpoints. The Performance Prism of Neely et al. [54] considers strategy, capabilities, and stakeholder views. Kaplan and Norton [39] have introduced the Balanced Score Card (BSC) including different perspectives like finance, customer, development potential and process perspective.

In conclusion, what all these measurement systems have in common is the focus on KPIs ignoring soft goals, meaning of maturity, relations, and dependencies. Due to the focus on comprised qualitative and quantitative KPIs, an assessment of additional process success factors is not enabled and complex interdependencies, human interactions and capabilities are not realized assessing the performance driver of a process.

The assessment of process performance drivers is seen as important step to gain knowledge of processes and increase understanding of performance and results. In this context, visibility to quantify qualitative parameters is seen as a key to increase knowledge of processes and to bridge the gap between strategic management and process level. Visibility is defined as the ability, usefulness, trust, and accuracy to exchange information [53]. Information is essential to reduce uncertainty and to increase knowledge about processes [7], [60]. Based on these considerations, the goal of the paper is to design a reference model (Performance Assessment System, PAS) to enhance process assessment in order to better gain information on business process stakeholders and performance.

Measuring processes is the key to give the management information, helping to reduce uncertainty and to take appropriate decisions [61], but measurement replaces trust and confidence with compliance and assurance. It enables auditing mechanisms by delegation of control and use of certification systems in control chains [47]. The paradigm of measurement has pervaded the business domain and organizations that are not in focus of audit and control [48].

PMS address the measurement of effectiveness and efficiency of business processes. However, PMS focus on numeric KPI, which do not foster the assessment of complex

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performance interdependencies and sophisticated measures, like goals or critical success factors. Instead, PMS reduces process performance to rituals of verification that inform the management ambiguously [47]. Empirical findings of Technical University of Vienna [30] and BP&O [37] show that companies use management methods like Continuous Improvement Process, BSC, Six Sigma, and Lean Management to enhance process performance. The effects of Lean Management e.g. are being measured in pieces of work in progress (WIP), cycle time, cost, quality and revenue aligned on targets and milestones [38]. The six sigma methodology uses measurement in step two of the DMAICR method of define, measure, analyze, improve, control and report [44]. Within the method of continuous improvement [45], measurement is hidden in step three of the well-known Deming cycle: plan, do, check and act. Moreover, Deming states that mechanisms of improvement can better be visible through statistics [43], thus pointing to common statistical process control (SPC) methods which presume numeric values.

Reviewing the literature, we have identified four different categories classifying the PMS: First, KPI that build up numeric process parameters to measure the efficiency of processes. Second, numeric process metrics consider the effectiveness of processes. Third, verbal descriptions provide information to evaluate the process efficiency. Finally, verbal descriptions that deal with verbal evaluation of process effectiveness [35]. In conclusion, current used PMS methods refer to the first and second category with a strong numeric focus. For this reason, complex interdependencies are reduced to single parameters [31]. Otherwise, qualitative success-factors are circumscribed using a whole range of measures [1], [27], [55] or totally ignored [41]. The result is a simplified assessment and reduced meaning of complex qualitative key-factors. Due to the numeric paradigm, success factors that have a qualitative nature, that are intangible and hard-measurable, lead to a reduced meaning [1], [31], [32]. Reduction in meaning is one obstacle to a successful implementation of PMS.

Qualitative performance problems remain fuzzy since they are hardly or not visible at all to numeric parameters [50] despite empirical studies show that a better visibility of business performance is demanded [46]. Although the concept of visibility is mainly used within the Supply Chain Management, we think that visibility is essential to highlight non-numeric key-factors. Once again, it is important that the management get all appropriate information renouncing reduced complexity and facilitated meaning.

In order to improve the knowledge about processes and their performance, it is the contribution of this paper to design a concept that enables visibility and improve business performance measurement and to foster knowledge about processes.

To reach our research goal, the remainder of the paper is as follows: Section 2 describes the universe of discourse. Based on an in-depth literature review, it presents the drawbacks of current PMS in respect to indirect performance indicators. Following the literature review, the research design contained in Section 3 consists of a classification of performance indicators, the derivation of the model, the description of the indicators and its application in a case study. Section 4 discusses the results and implications for further research.

II. Problem Statement

CIOs have continuously identified process improvement and advanced next generation analytics as important strategic priority [6], [5]. The need to conduct BPM with strategy is emphasized by various empirical studies, e.g. [46]. Improving business processes require the right information to identify, analyze, and re-design business processes. The management uses various methods of PMS to get information, especially Benchmarking, the Balanced Score Card, Continuous Improvement, Total Quality Management, and Six Sigma [30]. At the same time, empirical studies reveal a gap between process analysis, implementation, and execution [25]. Only a minority of the surveyed companies sees a correspondence between strategy and BPM [29]. One reason is the lack of methodological guidance of PMS due to a strong orientation of numeric key-indicators [49].

Although measurement is considered as the basis of control, compliance, and assurance [48], [47], it is strongly related to numeric measures. Qualitative performance problems, especially non-numeric key-performance indicators as well as processes with a qualitative-related output lead to fuzzy measures that are hardly or not visible at all to numeric parameters [50]. Therefore, qualitative measurements might be underrepresented or even left out to measurement. Complex problems are reduced to single parameters [31] and circumscribed with various range of measures [1], [27], [55]. Extra cost for finding, defining, applying, and maintaining measures [1], [32], [33] often lead to an ignorance of qualitative measures in PMS [41]. For this reason, qualitative performance indicators are invisible. Ignorance and reduction of measurements are one reason that important information is not available for the management.

Though, getting the right information is essential to reduce uncertainty and take appropriate decisions [47]. Even qualitative performance parameters have a meaning to improve business process performance. If processes with qualitative backgrounds are not assessed at all or are hard to catch via circumscribing measures, performance problems stay invisible. The result is a reduced comprehension and visibility of performance key-figures that consolidate the gap between strategy and business processes.

Therefore, we believe that visibility of qualitative and indirect performance key figures will improve the comprehension of business process performance indicators and increase information supply for the management. According to Caridi et al. [53], visibility is a term that is related to Supply Chain Management. In this context, they have developed indicators from collected features and measures of state-of-the-art supply chains in a three-step-procedural model:

1. Classification of the main types of information, which can be exchanged,
2. Definition of the most important features of the information flows,
3. Development of a set of metrics and a measurement model to assess the level of supply chain visibility quantitatively.

In analogy, we define visibility of process performance as the principal ability to assess process performance information. It is established by using a performance assessment system that gives appropriate information through numeric and verbal performance descriptions.

Therefore, the question arises, if visibility will reveal additional process performance information and will foster a more meaningful understanding of processes. Based on these considerations, it is the contribution to design a concept as reference model considering visibility to improve assessment of qualitative and verbal-performance indicators. To do so, we apply the above mentioned procedural model to state-of-the-art PMS. The contribution finally follows the research question whether visibility improves information and fosters understanding and knowledge in order to improve the correspondence between strategic level and BPM.

III. LITERATURE REVIEW

In order to show that current used PMS do not consider a comprehensive measurement of qualitative and indirect performance indicators (the goal of a state-of-the art according to the principles of Fettke et al. [51]), we apply a systematic literature review that analyzes relevant work as well as current findings. Based on a review scheme of Denyer et al. [28], the review considers several databases, which are included in the AIS ranking list and the German WKWI ranking. We have applied a key-word list that considers relevant terms of PMS, like KPI, quality, measurement, non-numeric, or verbal. By using a forward and backward search, 60 articles have been identified that consider the research question.

The first step of the procedural model is the classification of the main types of information. We were able to categorize the relevant models found during the review by key elements related to the information type domain, scope and measures by which the context and approach to the model is driven predominantly.

Table 1 contains an overview of relevant evaluated models with regards of their originating domains, scope to process assessment, and recommended measures.

Table 1. Main types of information of PMS

Author	Domain	Scope	Measures
Aburub et al. [57]	Goals		Goals, Measures
Almeida et al. [24]	Soft Goals	Accessibility, Confidentiality, Completeness, Accuracy, Traceability, Integrability, Trust and confidence, Emphaty	
Andersen [15]	Productivity	Productivity, Effectiveness, Intangible dimensions	KPI
Becker et al. [20]	Process Modelling	Definition of process building blocks that abstract and describe	domain specific language repository in an ontology
Brewer et al. [14]	Logistics		Internal process measures, Intangible measures
Bruce et al. [11]	Benchmarking		Stability, Skills, Staffing, Automation, Technology
Cardoso et al. [26]	EPC Quality		Loops, Parallel paths, Joins, Splits
DeToni et al. [10]	KPI	Cost/Productivity, Time, Quality, Flexibility	
Heinrich et al. [58]	BPMN	Semantic annotation of parameters for business process quality	Suitability, Maturity, Operability, Effectiveness, Productivity, Compliance, Variants, Components, Media disruptions, Capacity, Capability
Juan et al. [17]	EPC Quality		Scenario Similarity Degree
List et al. [19]	Performance Measurement		Instances, Cycle Time, Revisions, Complaints
Moxham [13]	Non-Profit organisations	Political reasons, Organizational transparency, Community involvement	Public confidence, Staff quality
Naumann et al. [16]	Information Quality	User, Source, Process	
Pavlovski et al. [22]	Process Modelling	Extension of BPMN with certain describing parameters	operating condition, control case
Popova et al. [12]	Performance Indicator	Process, Performance, Organization, Agent	Discrete or continouous, Assessable with a scale, Qualitative soft or Quantitive measurable, Customer satisfaction, Company reputation, Employee motivation, Hard or soft goals, Process implicit indicators
Remus [21]	Knowledge Management	Time, Cost, Quality, Knowledge intensive processes	Complexity, Variability, Granularity, Competence
Stein et. al [42]	ARIS	annotation of ontologies	
Wettstein et al. [18]	PMS Maturity	Measurement, Data collection, Storage of data, Communication of performance results, Use of performance results, Quality of performance measurement process	Maturity
Vanderfeesten et al. [59]	EPC Quality		Coupling, Cohesion, Complexity, Size

The second step demands the definition of the most important features of information flows. To do so, we interpreted the aforementioned different main types, scope and measures as main drivers of the models and put them into context to the classification of the performance systems defined in [35] and referred to in Section 1. This classification distinguishes models for process assessment that use numeric or descriptive features as well as address efficiency or effectiveness. So, we could describe the important features of the models through their effect in the performance systems and hence, their contributions to visibility. These relations are depicted in a reference model catalog (RMC). In it, the main driver behind the model is used to describe the models by the classifier domain and its original and working domain is described in the subdomain section. The access part displays the effect of the model in the respective performance systems.

Table 2. Reference model catalog for PMS vs. performance systems

Classification		Main Part		Access Part			
				Process Success / Efficiency		Process Outcome / Effective-ness	
Domain	Subdomain	No	Author	Parameters	Factor	Parameters	Factor
Measure driven	Productivity	1	Andersen [15]	X		X	
	Logistics	2	Brewer et al. [14]			X	
	Benchmarking	3	Bruce et al. [11]			X	
	Goals	4	Aburub et al. [61]		X		
	EPC Quality	5	Juan et al. [17]			X	
		6	Cardoso et al. [26]			X	
		7	Vander-feesten et al. [60]			X	
	No restriction	8	List et al. [19]			X	
		9	Popova et al. [12]	X	X	X	
		10	Heinrich et al. [58]			X	X
Scope driven	Knowledge management	11	Remus [21]	X		X	
	Information Quality	12	Naumann et al. [16]			X	X
	Soft Goals	13	Cardoso et al. [24]		X		X
	Non-profit-organizations	14	Moxham [13]		X		X
	PMS Maturity	15	Wettstein et al. [18]			X	X
	No restriction	16	DeToni et al. [10]	X			
		17	Pavlovski et al. [22]				X
		18	Becker et al. [20]		X		X
		19	Stein et. al [42]				X

The RMC also reveals that there is no model solution that features impact and visibility in all performance systems or domains. So, a restricted visibility can be assumed not only in case of using conventional PMS that rely on numerical KPIs, but as well on using these exhibited models that omit important features of information flows. Hence, information on processes and their performance will stay invisible or restricted to their working domains.

This highlights the lack of a certain combined or integrated model for business process assessment that enables full visibility independent of business domains that will contribute to better information on process performance for the management.

IV. Research Design

The research process results in a reference model, which bases on definition of Schütte [8]. As representation of real world phenomena [36], a reference model is used for the design of organizations and information systems and serves as a recommendation [3]. In order to enable visibility and foster process knowledge, the resulting model builds on deductive and inductive elements. Whereas the empirical foundation is constituted in a systematic and scientific literature review (see Section 2), the theoretic construct of visibility is discussed in further work of Caridi et al. [53]. Building the model orientates on the guidelines for reference modeling of Becker et al. [3], Schütte [8] and Frank [23].

As result, the research design orientates on the Design Science approach of Hevner et al. [2]. The empirical examination of PMS in respect to visibility of indirect indicators builds only an initial step. In order to get a reference character, the design model has to be implemented, evaluated, and tested. For this reason, the results of the model are applied and discussed in a case study in a public company.

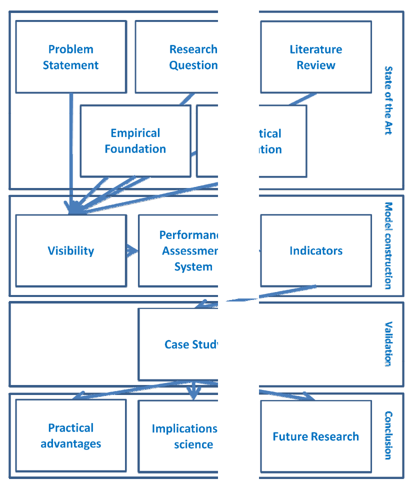

Figure 1. Research Design

A. *Model construction*

In order to develop a Performance Assessment System (PAS) that is able to deliver comprehensive visibility, we interpreted, aggregated and abstracted the above mentioned main types of information (cf. Table 2) and matched most important features. The third step of the procedural model is to abstract and develop measures as well as a model that finally enables best visibility of process performance from these features.

In the case of numeric values, the consequent distinction between the scopes of process success (efficiency) and outcome (effectiveness) leads to a distinct categorization. E.g. quality as a popular term found in the review is an indicator that has many definitions and meanings in a lot of contexts, but it always points to a certain intrinsic sphere of a case. As process related concept, it also can be described as measure of how effectively a process meets the customer's needs (by means of process results (Schneiderman [4]) or as capability to meet the customer's requirements under all possible constraints (Actano [40]). On descriptive models, we used the distinction between the application of direct goals that indicate success or failure of a process and a broader application-driven view to the semantic annotation of other existing models, e.g. BPMN besides the use of process related ontologies [35] as such.

So, four generic indicators can be derived as measures that finally form the PAS reference model (Four-Box-Model). They are displayed in Table 3 and discussed in detail in the following section.

Table 3. Contents of PMS, derivation of generic indicators and the Four-Box-Model (grey)

Performance System	Contents		Generic Indicator Name	Short form
Parameters for process efficiency	Numeric parameters that point to process success, e.g. measures or KPIs		Key Performance Indicators	KPI
Factors for process efficiency	Hard or soft goals, success factors	→	Process Success Factors	PSF
Parameters for process effectiveness	Parameters pointing to quality, internal measures, skills, etc.		Process Metrics	PMX
Factors for process effectiveness	Semantics, e.g. ontologies or semantic annotation		Process Ontology	PO

B. Indicator properties

1) Key Performance Indicators

For a lot of performance problems especially in production or processes directly related to company success or value, KPIs might be sufficient and are undoubtedly a good starting point for assessment. As there are quite a lot of other discussions, recommendations and solutions available on KPIs already, we will not collect or compare individual indicators here but point to existing catalogs like e.g. Eckerson [62] or Baroudi [63]. Most common KPIs are financial-related like e.g. Return on Investment (ROI), Cash Flow, Earnings before interest and tax (EBIT) etc. Increasing interest is being given to non-financial KPIs in the past. An example for a collection of production-related KPIs is the German Engineering Federation (VDMA) industry norm that defines and finally standardizes branch-independent production KPIs like throughput or productivity as well as static indicators like scrap rate or process capability [67].

There are several distinct reasons for using KPIs particularly for the assessment of production processes:

1. Processes in industrial manufacturing are usually at systematized and automated. This pre-structuring facilitates the usage or derivation of (intermediate-) parameters, e.g. through machine hours, throughput figures or downtime,
2. Results of production processes are often formulated in appropriate numeric parameters direct relevant to the income and hence, to money (e.g. units, yield, selling price)
3. The products manufactured normally embody the biggest part of the goods that contribute to the enterprise revenue. So changes to the indicators, e.g. increasing throughput immediately visually and countably impacts the operating result.

2) Process Success Factors

Via an intermediate step of paraphrasing an abstract problem into a goal or so-called process success factor (PSF), the success of the process can be rated transparently. They can be considered as conditions that are composed in free text and define a process success when fulfilled; the result of the process is basically rated, not directly evaluated with a parameter. The factors can take shape of either a binary yes-no decision or a discrete cascade of opinions about the process success. The opinion cascade can be a set of descriptive statements or quantized quasi-values. Examples for quantifications are

1. verbally escalating statements indicating increasing value like "attended the French class", "overtook the business correspondence in French" or "is responsible secretary for conferences in foreign languages" as target corridor for the PSF "the employee enhances his language knowledge",
2. rating through a describing multi-level system like not, rudimentary, partly, mostly or fully archived,
3. additional application of a score range to this cascade (e.g. 1-5) similar to school grades, evaluation from 0-100% or application of a finite point scoring scale of a parameter (e.g. 20 points maximum).

An advantage is that PSF can be quantified with a numeric quasi-value after a mandatory declaration and paraphrasing step before. This improves IS-supported processing, display, and control similar to KPIs and leads to a better visibility and comparability of concerned performance problems.

3) Process Metrics

Many business processes, even most of the support processes cannot be assessed via the validation or rating of the process success, but by evaluating process effectiveness by means of the process featuring a certain outcome and is performing at all. Hence, they are immanent parameters that describe the processes' formal building blocks or what it takes to execute a certain task. The so defined indicator is called Process Metrics (PMX) and can be applied both to single process parts or the entire business process. Examples of individual indicators out of this family could be abstracted and logically derived from the comparative analysis earlier in this paper, e.g.

1. Interruptions are the amount of media breaks in data or documents to be processed along the process chain,
2. Responsibles count persons or systems that are executing a step,
3. Approvers count persons or logic components that have to authorize or approve the work done in a step,
4. Complexity is the subjective degree of difficulty to execute a step or
5. Knowledge is the subjective degree of the responsible's ability to execute a step.

Process Metrics can be directly countable or indirectly rated and hence quasi-valued, quantitative or qualitative. Combinations of parameters can be used to better assess outcome through the entire process landscape, e.g. the ratio of knowledge of a responsible (person) compared to the complexity of the step, or a product of amounts of involved approvers by departments.

4) Process Ontology

In order to prepare a PO for an indicative purpose, its construction should contain at least enterprise-specific elements and such that emphasize the performance, relation and time orientation structure of business processes like

1. an explicit relation that determines a logical predecessor or successor, either mandatory or optional in order to model and ensure a certain sequence,
2. role-related properties that declare roles and responsibilities, e.g. with a RACI model (Smith and Erwin [64]) and
3. performance-related properties (indicators)

that can be attached to statements of their rating, either verbal or numeric. These properties delimit the concept of the PO also from other (formal) taxonomies like KPIs.

By definition, transparency of domains will increase simply by collating data and relations to fill in the ontology. Regrettably, direct insights might rather stay restricted to the persons developing this indicator since the use of semantics in management is still not very common (Merdan et al. [65]), and we will partly withdraw from the idea as using it as an indicator as proposed in [35]. Though, we will keep the wording "PO" in the Four-Box-Model since the use of semantic techniques as such usually is associated with ontologies.

In contrary, semantics can be applied by adding descriptive features of process outcome e.g. to state-of-the-art business process management systems (BPMS) like process models. Through this enrichment of the model content with descriptive performance information (e.g. roles, goals), visibility also is increased. BPMS today only include the possibility of annotating numeric indicators very rudimentary (e.g. amount, time and cost of steps), and not in our sense of a comprehensive approach to visibility of information. So, the PAS can be considered as a complementing feature for BPMS in case its contents are annotated and hence, information on processes and performance is visible for the stakeholders. In this context, the PO requirements would be addressed already through the design of the BPMS system without the redesign of an ontology as a carrier.

C. Case Study

Yin [66] describes a case study as "an empirical inquiry that investigates a contemporary phenomenon within its real-life context, especially when the boundaries between phenomenon and context are not clearly evident". It is used to analyze and describe complex cause and effect chains in situ to explain sophisticated problems. It is also the method of choice if phenomena cannot reasonably be decomposed into smaller objects of investigation. This was the case in a project that was outlined to assess the administration processes in a German university during the year 2011. There, the PAS was introduced in around fifty workshops for process analysis and optimization, and we intended to already catch the results of the user acceptance in a case study though the workshop design and execution was still evolving. The originally intended assessment method for the processes was Six Sigma, but since processes in public administration and in particular in higher education, e.g. exam management

1. are rather of qualitative nature and so, are not easily assessable by numeric values and
2. the statistically reproducible success of the process is not questioned,

we additionally used the Four-Box-Model as assessment method. On the one hand, it allowed assessing internal performance (effectiveness) by using PMX, e.g. the amount of interruptions. While project conceptualization, we agreed to use a Lean management approach to detect waste in the process rather to compile and compute measures, and so we could use PSF as indicators that described a desired result rather than evaluating it, e.g. compliance with examination regulations.

We proceeded as follows: In case the success or the outcome of a process could not easily be described by KPIs, we determined the scope of the assessment and either tried PSF for the description of success or PMX for the evaluation of intrinsic features as next successive indicator. By interpreting the results of the assessment, first entry points to optimization could be identified. The participants of the workshops vastly agreed to a perceived usefulness of the approach as numerous performance problems and waste became visible that were hidden in quantitative and therefore, previously untouched processes. The identified performance problems were addressed to the process owners and process improvements were put in place in order to fulfill the desired results derived from the actual results of the assessment. As the project is not yet finished, we consider the validation process as still in progress. It will have to be validated further by

1. broadening the application to other companies and
2. performing a structured survey

pointing to the technology acceptance model (TAM). But the principal applicability of the PAS Four-Box-Model in business process management can be considered to be given.

D. *Findings*

Using the research design to collect and apply key elements contained in reviewed models, additional available assessment methods could be identified and expanded from numeric to descriptive indicators. Through the design according of the three steps of the procedural model, a concept of visibility of business processes performance could be introduced and defined.

Finally, a combined Four-Box-Model for performance assessment of business processes could be deployed that contains a broad range of possibilities to identify performance problems and to assess performance beyond KPIs. In the case of process ontologies, transparency is even more likely to increase when its working principle of semantics is used to integrate various indicators rather than only to represent one. Possibilities to do so are e.g. by simply adding semantic annotation to a process notation model that users might have in place already as a carrier. On the long run, the annotation of the whole PAS in the form of an ontology would be preferable.

In a first business case study, the applicability and usability of the PAS was approved by the users' subjective perception of an enhanced visibility of process problems which is a promising signal for future application. Though, the usefulness still has to be further augmented in a rigorous way.

V. Conclusion and Outlook

The Four-Box-Model as Performance Assessment System for business processes proposed in this contribution incorporates four different viewpoints to business performance problems and broadens assessment possibilities from numeric KPIs to three additional indicators. It is able to identify process problems that are hard-measurable or invisible to KPIs by evaluation and rating as well as consideration of efficiency and effectiveness.

Moreover, it is able to assess the concealed performance potential beyond the boundaries of using KPIs through enhanced visibility. Though, most results of the assessment can also be formulated in figures, thus is still leaving room for the implementation in popular dashboards. The indicators contained in the model can be used individually for isolated performance problems, but thorough assessment should be performed by a consecutive application of all indicators.

Due to its initial design and openness, it incorporates all business processes, not only production or value creating processes. It contributes to a better visibility of information for the management in order to control operative and strategic goals of a company. Nevertheless, it is ready-to-use for enterprises' business process assessment immediately and without the use of a certain framework, programming or modeling language.

A contribution to the research community is the concept and discussion of combining or integrating various different approaches for assessing business processes. Especially the use of semantic technology to integrate the solution opens room for discussions on implications of the reference model. Direct annotation of the presented indicators to process notation models would already add value to them and enhance their usability and relevance in the future. More than that, the presented model could also be integrated in a semantic framework which additionally raises the possibilities of process assessment.

Hence, continuing research in the future will be on additional case studies to validate the applicability and usability of the model and how to integrate the given indicators into an ontology that subsequently is annotated to a process notation model.

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